Battery Supply Chain Strategy

This memo evaluates whether an electric-vehicle startup should vertically integrate its battery supply chain. The classical argument is well established in the strategy literature.

Vertical integration is an arrangement in which the supply chain of a company is integrated and owned by that company. Usually each member of the supply chain produces a different product or market-specific service, and the products combine to satisfy a common need. When a company expands its business into areas that are at different points on the same production path, it is said to be vertically integrated.

Backward integration involves acquiring or controlling suppliers of raw materials and components, while forward integration moves toward distribution and the end customer. For an EV manufacturer, cell production is the obvious backward target because cells represent thirty to forty percent of vehicle cost.

However, the textbook case understates two modern complications. First, cell chemistry is still evolving, so owning a gigafactory locks the firm into a technology that may be leapfrogged within five years. Second, capital intensity crowds out investment in the charging and software layers where differentiation is actually visible to consumers.

Our recommendation is a hybrid: take minority stakes in two cell suppliers with options to expand, while keeping final pack assembly in-house for quality control. This captures most of the cost advantage of ownership while preserving technological flexibility.